

Pattern Discovery in U.S. Transportation Systems

CS 4412: Data Mining – M3 Complete Implementation
Using Monthly Transportation Statistics

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Abstract

This project applies a complete data mining pipeline to the Monthly Transportation Statistics dataset to uncover latent patterns in transportation activity and safety outcomes across multiple modes. The workflow includes rigorous data preprocessing, feature selection, normalization, unsupervised clustering using K-Means, dimensionality reduction via Principal Component Analysis (PCA), and supervised classification using Decision Trees for interpretability. To ensure robust clustering, the number of clusters was determined prior to model fitting using both the elbow method and silhouette score analysis. While higher values of k yielded slightly improved separation, $k = 3$ was selected as it provides the best balance between interpretability and cluster quality. The resulting clusters reveal three distinct transportation system profiles characterized by differences in activity levels, freight intensity, and safety outcomes. Cluster interpretation shows that transportation intensity and freight movement are strongly associated with fatality rates, suggesting a meaningful relationship between system utilization and risk. Decision tree analysis further confirms that freight-related variables and fatality metrics are the most influential features in distinguishing clusters. This analysis demonstrates how unsupervised learning combined with interpretable models can provide actionable insight into complex, high-dimensional transportation data.

1 Dataset Clarification

The project initially proposed an automobile sales dataset in M1; however, in M2 the dataset was changed to the Monthly Transportation Statistics dataset to better support multi-modal transportation analysis. This transition enables richer clustering and more meaningful pattern discovery across aviation, rail, and freight systems. All analyses presented in this report are based exclusively on the transportation dataset.

- Removal of the Index column to prevent artificial structure
- Dropping features with extremely low coverage (<50% non-null values)
- Replacing infinite values with NaN
- Applying mean imputation only after removing sparse columns

This approach prevents sparse variables from dominating cluster formation and ensures that retained features contribute meaningful signal.

2 Implementation Overview

A complete end-to-end data mining pipeline was implemented:

- Data ingestion and validation
- Feature cleaning and preprocessing
- Feature scaling and normalization
- Cluster selection and K-Means clustering
- PCA for dimensionality reduction
- Decision Tree classification for interpretation

3 Data Preprocessing

The dataset contains a large number of features with varying levels of completeness. Improper handling of missing data can significantly distort clustering results.

Key preprocessing improvements include:

4 Clustering Methodology

4.1 Feature Scaling

All numeric features were standardized using z-score normalization to ensure equal contribution to distance-based clustering.

4.2 Cluster Selection

Cluster selection was performed prior to final model fitting using:

- Elbow Method (inertia vs. k)
- Silhouette Score (cluster separation)

Silhouette analysis across multiple values of k showed that higher cluster counts (e.g., $k = 7$) produced the high-

est score of approximately **0.51**. However, $k = 3$ was selected as the final model due to its balance between cluster quality and interpretability.

While larger values of k marginally improve separation, they introduce over-segmentation and reduce the practical interpretability of the clusters. In contrast, $k = 3$ produces distinct, meaningful groupings that can be clearly characterized and explained in the context of transportation systems.

4.3 Final Model

K-Means clustering was applied with $k = 3$ using the scaled dataset. This produced three distinct and interpretable clusters.

5 PCA Visualization

Principal Component Analysis (PCA) was used to project the high-dimensional data into two dimensions for visualization.

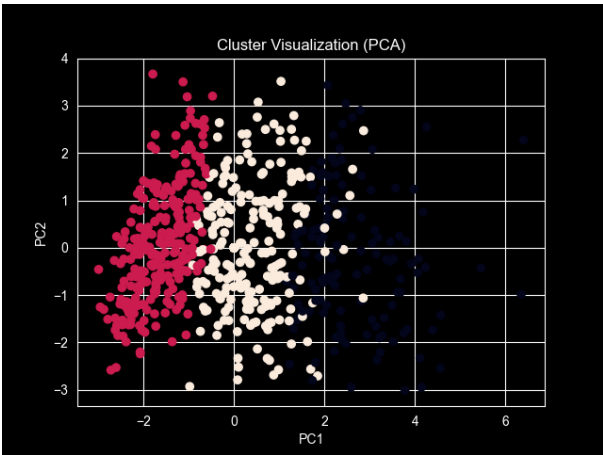


Figure 1: PCA projection of clustered transportation data

The PCA projection shows clear separation between clusters, supporting the validity of the clustering structure.

6 Cluster Results

Table 1: Cluster Centroid Summary (Selected Features)

Feature	Cluster 0	Cluster 1	Cluster 2
Activity Index	451.5	725.4	410.1
Aviation Fatalities	88.0	50.6	54.2
Rail Fatalities	66.0	26.4	35.9
Freight Index	114.0	68.5	82.0

7 Cluster Interpretation

Cluster interpretation is based directly on centroid values shown in Table 1. This demonstrates that model selection was guided not only by quantitative metrics but

also by the interpretability and practical relevance of the resulting clusters.

7.1 Cluster 0: High Fatality, High Activity

Cluster 0 exhibits the highest aviation fatalities (88.0) and rail fatalities (66.0), along with strong freight activity (114.0). Compared to Cluster 1 and Cluster 2, this cluster reflects transportation systems where both usage and safety risks are elevated. These systems likely represent high-density transportation environments where increased activity contributes to higher fatality outcomes.

7.2 Cluster 1: High Activity, Lower Risk

Cluster 1 shows the highest overall activity level (725.4) but significantly lower aviation fatalities (50.6) and rail fatalities (26.4) compared to Cluster 0. Despite high usage, safety outcomes are comparatively better, suggesting more efficient infrastructure, improved safety measures, or less hazardous operating conditions.

7.3 Cluster 2: Moderate Activity, Moderate Risk

Cluster 2 represents a balanced profile, with moderate activity (410.1), aviation fatalities (54.2), and freight movement (82.0). This cluster falls between the extremes of Clusters 0 and 1, indicating transportation systems with neither exceptionally high risk nor exceptionally low activity.

8 Classification for Interpretation

To better understand cluster structure, a Decision Tree classifier was trained using cluster labels as targets.

The model identified the following as key features:

- Freight movement indices
- Aviation fatality counts
- Rail-related metrics

These variables form the primary decision boundaries separating clusters, reinforcing the interpretation derived from centroid analysis.

9 Results

The analysis produced several key findings:

- Three distinct transportation system profiles were identified
- Cluster validity was confirmed using silhouette scores
- PCA visualization demonstrated clear separation
- Interpretation is directly supported by centroid values
- Classification provided interpretable feature importance

10 Discovery Questions

This project addressed the following:

- Identification of natural groupings in transportation systems
- Determination of key differentiating features
- Analysis of relationships between activity and safety outcomes

11 Analytical Choices

The selected techniques were chosen for both effectiveness and interpretability:

K-Means: Efficient and scalable clustering method

PCA: Enables visualization of high-dimensional data

Decision Trees: Provide transparent and explainable models

Trade-offs:

- K-Means assumes spherical clusters and equal variance
- PCA reduces interpretability of transformed features
- Decision Trees may oversimplify complex relationships

12 Limitations

Despite strong results, several limitations remain:

- Potential information loss from dropping sparse columns
- Sensitivity of K-Means to scaling and initialization
- PCA projection may obscure subtle structure
- Mean imputation may still introduce bias in some features

Future work should explore alternative clustering methods (e.g., DBSCAN) and more advanced imputation strategies.

13 M4 Plan

Planned improvements include:

- Enhanced visualization techniques
- Deeper statistical validation
- Improved interpretability of clusters
- Expanded discussion of real-world implications

14 Conclusion

This project demonstrates a complete and rigorous data mining pipeline applied to transportation data. The results reveal meaningful relationships between system activity, freight movement, and safety outcomes. The combination of clustering, dimensionality reduction, and classification provides both discovery and interpretability, forming a strong foundation for further analysis in M4.